Portfolio Risk Analysis

Portfolio: Equal-weighted 8-asset basket — AAPL, MSFT, GOOGL, JPM, BRK-B, GLD, TLT, SPY
Period analysed: Jan 2019 – Dec 2024 (1,508 trading days) **Portfolio value (assumed):** $1,000,000
Author: Pat Ploypairaoh | **Date:** 4/20/2026

Executive Summary

Over the 2019–2024 period this equal-weighted portfolio of four US equities, a value conglomerate, gold, long-duration treasuries, and the S&P 500 delivered 16.71% annualized return at 16.83% annualized volatility, yielding a Sharpe ratio of 0.725 against a 4.5% risk-free rate.

The headline risk figure: 1-day 99% Value-at-Risk is $29,146 under the historical method and $24,000 under the parametric (normal) method on a $1M portfolio. The $5,147 gap is direct evidence of fat-tail behaviour in the return distribution — real tail losses arrive more often and run deeper than a normal distribution predicts. On the 1% worst days, expected shortfall (CVaR) averages $43,829 under the historical method — roughly 50% larger than the VaR threshold alone.

Backtesting on a held-out 2024 test year shows the parametric 99% VaR was well-calibrated (2 actual breaches vs 2.5 expected, Kupiec p = 0.73, not rejected). The three other model-confidence combinations were rejected in the over-conservative direction — fewer breaches than expected — reflecting that the 2019–2023 training window contained substantial high-volatility episodes (COVID, the 2022 rate shock) that made models over-estimate risk when applied to the calmer 2024 test period.

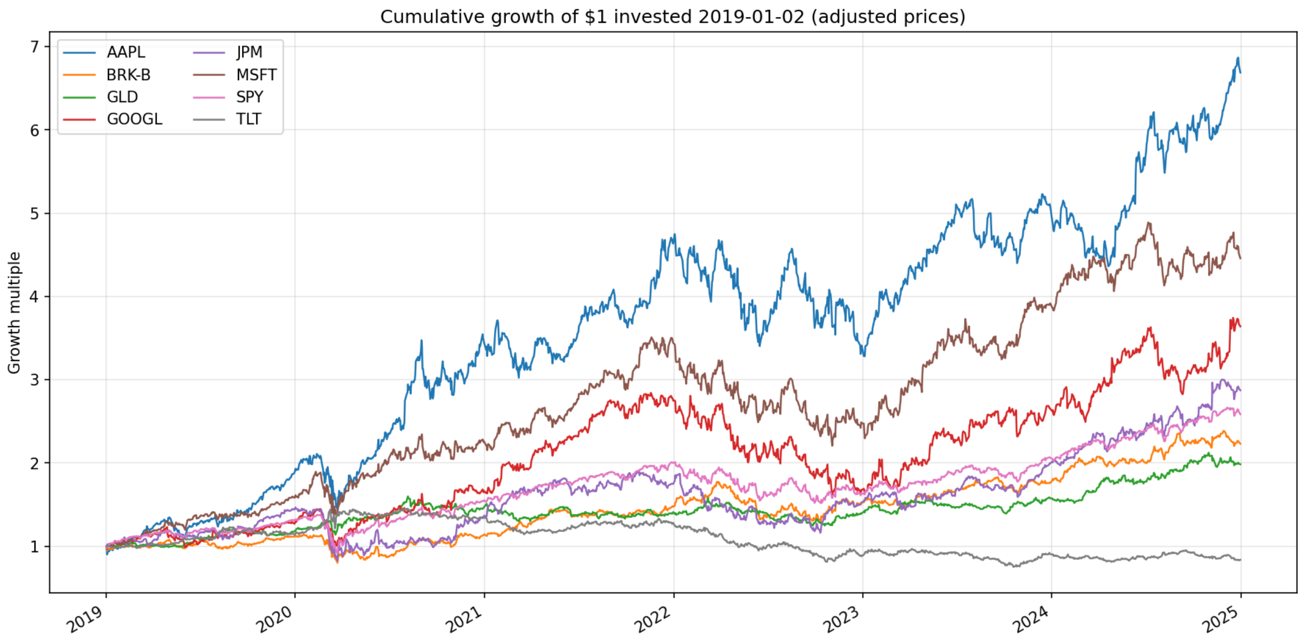
Recommendation: Implement the "moderate tilt" allocation (Section 6.2) — a 70% equal-weight / 30% max-Sharpe blend that raises the Sharpe ratio from 0.725 to 0.823, lifts expected return from 16.71% to 18.05%, reduces volatility from 16.83% to 16.45%, and lowers 99% VaR by 0.4%. This allocation strictly dominates the baseline on all four measured metrics while preserving meaningful exposure to every asset. Two alternative allocations — unconstrained max-Sharpe and minimum-variance — are presented in Sections 6.1 and 6.3 and are appropriate under different investment mandates.

1. Portfolio Composition

Ticker	Asset	Role	Weight
AAPL	Apple	Large-cap tech growth	12.5%
MSFT	Microsoft	Large-cap tech growth	12.5%
GOOGL	Alphabet	Tech / advertising	12.5%
JPM	JPMorgan Chase	Financials	12.5%

BRK-B	Berkshire Hathaway	Value conglomerate	12.5%
GLD	SPDR Gold Trust	Commodity (gold)	12.5%
TLT	iShares 20+ Year Treasury	Long-duration bonds	12.5%
SPY	SPDR S&P 500	Equity market benchmark	12.5%

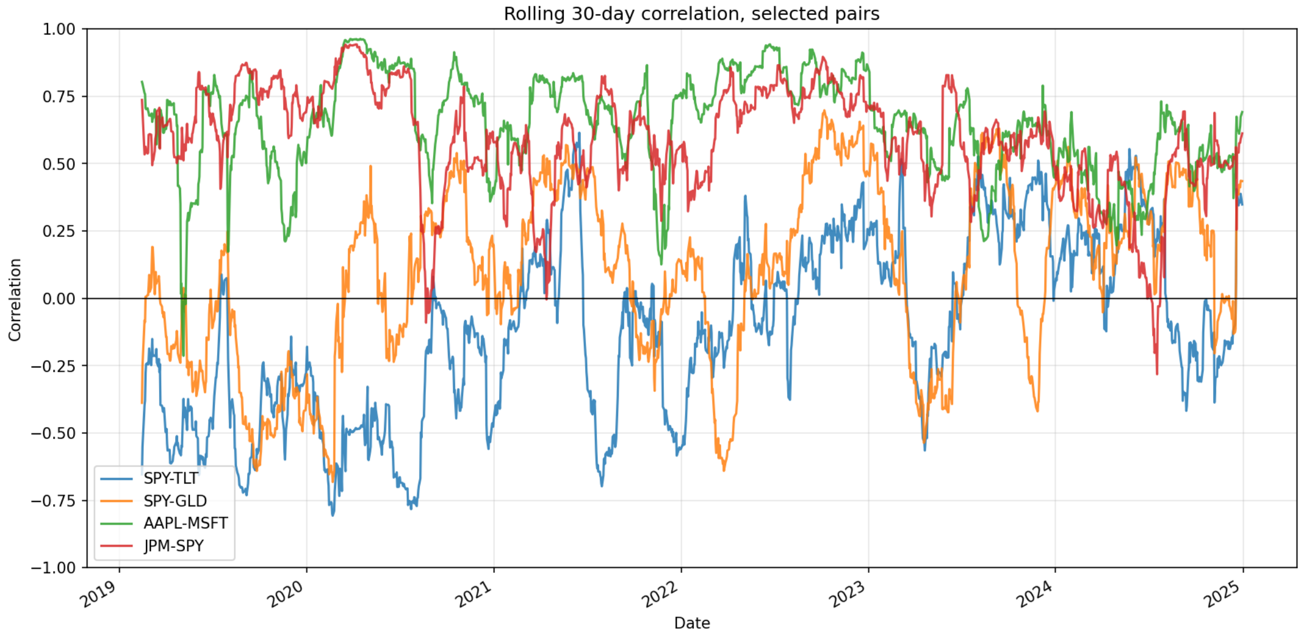
The portfolio was deliberately constructed to include three asset classes — equities, bonds, gold — to observe diversification effects.



2. Return & Volatility Profile

Metric	Value
Annualized return	16.71%
Annualized volatility	16.83%
Sharpe ratio (rf = 4.5%)	0.725

The rolling 30-day correlation analysis reveals that cross-asset correlations are not static. SPY-TLT correlation, typically negative (the "bonds hedge equities" regime), flipped positive through much of 2022 as aggressive rate hikes hurt both asset classes simultaneously. SPY-GLD swings between roughly –0.68 and +0.70 across the period. **This is the most important caveat attaching to the parametric VaR results below** — the parametric model assumes a fixed covariance matrix that cannot capture these regime shifts.



3. Value-at-Risk — Three Methods Compared

All figures below assume a $1,000,000 portfolio; multiply the return figure by portfolio value to dollar-ize. 1-day horizon.

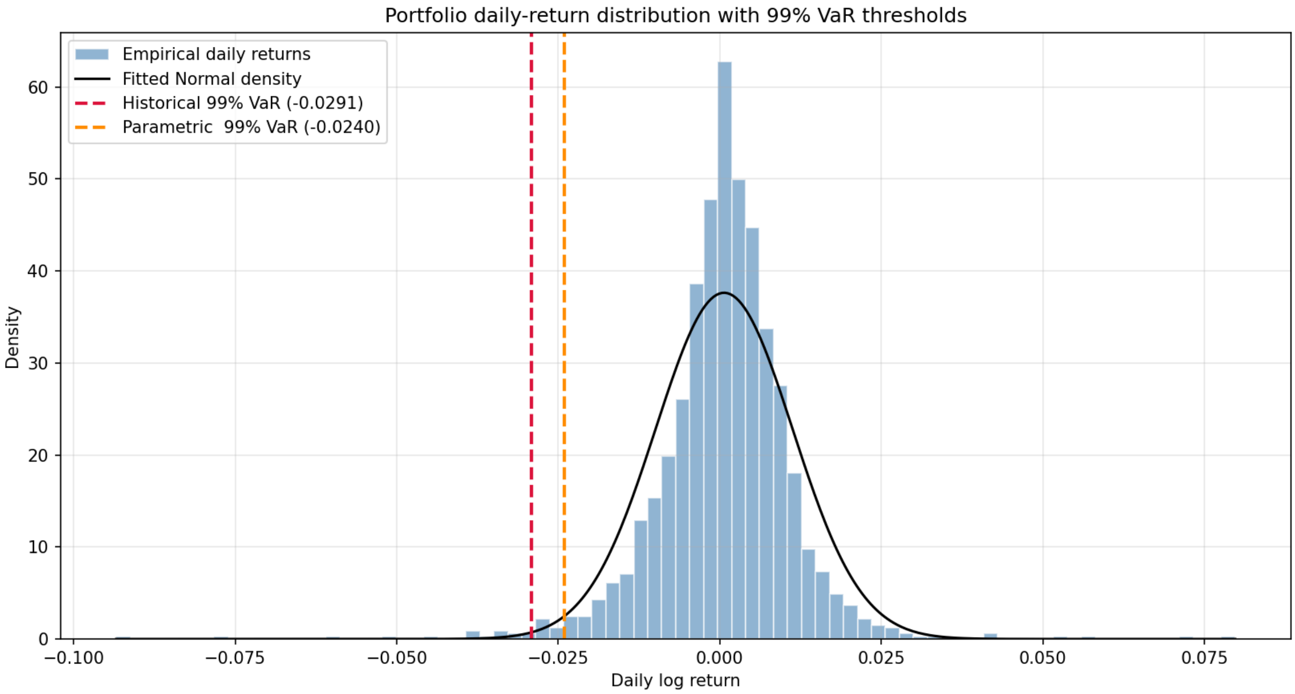
Method	95% VaR	95% CVaR	99% VaR	99% CVaR
Historical simulation	$15,531	$25,284	**$29,146**	**$43,829**
Parametric (normal)	$16,775	$21,205	**$24,000**	$27,592
Monte Carlo (10,000 paths, Cholesky, seed=42)	$16,689	$21,022	**$24,017**	$27,734

Key observation — the historical-parametric gap. Historical 99% VaR exceeds parametric 99% VaR by $5,147. Under a strict normal-distribution assumption, this gap should not exist. It does exist because real daily returns have fatter tails than the normal distribution — extreme losses occur more often and run deeper than the bell-curve predicts. The March 2020 COVID crash is the most visible instance of this in the 2019–2024 window, and the historical method captures it directly while the parametric method does not.

Monte Carlo agrees with parametric, as expected. The MC 99% VaR ($24,017) is $17 away from the parametric 99% VaR ($24,000) — sampling noise on 10,000 paths. This is the correct result, because both methods draw from the same multivariate-normal distribution; Monte Carlo simply does it via simulation rather than closed form. If the two had disagreed materially, it would indicate a bug. The real comparison that matters is historical-versus-parametric/MC — and that is the fat-tail story.

CVaR is a larger number than VaR at every confidence level, as it should be by construction. The historical 99% CVaR of $43,829 is 50% larger than the historical 99% VaR of $29,146 — meaning that on the days the portfolio did breach VaR, the average loss was half again as large as the threshold itself. This is the kind of figure Basel III expects banks to capitalize against, rather than VaR alone.

Diagnostic: Cornish-Fisher modified VaR. A fourth method was computed as a fat-tail diagnostic. Cornish-Fisher takes the parametric framework and corrects the Normal z-score for the empirical skewness and excess kurtosis of the return series. With sample skewness of –0.54 and excess kurtosis of +11.6, the corrected 99% VaR comes out at $55,856 — substantially *above* the historical figure. This overshoot is not a recommended risk estimate; it is itself the headline finding. The Cornish-Fisher correction is a Taylor-series approximation around the Normal distribution and is reliable for mildly fat-tailed returns, but past roughly K ≈ 6 the cubic moment term begins to dominate. The fact that the correction blows up here means the return distribution is so far from Normal that a moment-based adjustment cannot recover the true tail — a stronger version of the same finding that the historical-parametric gap delivers. The operational implication is unchanged: trust the historical and Monte Carlo numbers for the tail, not parametric.



3.1 Diversification benefit, quantified

A direct counterfactual test isolates the contribution of GLD and TLT — the two non-equity diversifiers in the portfolio. The same equal-weight construction was rebuilt using only the six equity assets (AAPL, BRK-B, GOOGL, JPM, MSFT, SPY) at 1/6 each, then compared to the actual 1/8 portfolio:

Portfolio	Annualized return	Annualized volatility	99% VaR	99% CVaR

6-asset (equities only)	20.87%	22.40%	$38,939	$57,353
8-asset (with GLD + TLT)	**16.71%**	**16.82%**	**$29,146**	**$43,829**
Effect of adding GLD + TLT	–4.16 ppt	–5.58 ppt	**–25.1%**	**–23.6%**

Adding GLD and TLT cut 99% VaR by roughly $9,800 (25%) and 99% CVaR by a similar magnitude, at a cost of 4.16 percentage points of annualized return. On a risk-adjusted basis this is a clear win — volatility falls faster than return, and Sharpe improves. The mechanism is in the correlations: TLT correlates –0.20 with the equity-only book over 2019–2024, and GLD correlates +0.08 (essentially zero). Neither is a strong hedge, but both contribute uncorrelated variance.

The caveat. The diversification benefit is regime-dependent. The rolling-correlation analysis in Section 2 documents that SPY-TLT flipped positive through much of 2022 as both asset classes sold off into rate hikes; a 2022-style joint sell-off is precisely the regime where this 25% VaR reduction would not be available. The honest framing is: *adding GLD + TLT cut full-period 99% VaR by 25% at a cost of 4.16 ppt return — but the reduction is averaged over regimes in which the diversifiers worked very differently, and the joint sell-off of 2022 remains the live tail risk this allocation is most exposed to.*

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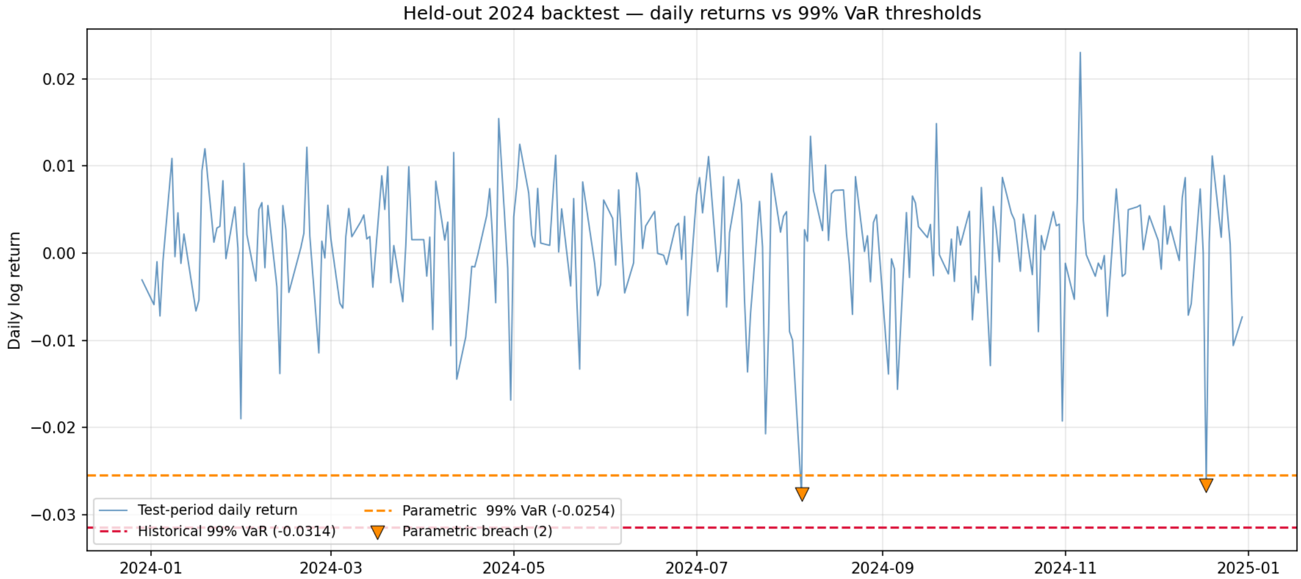
4. Backtesting — Kupiec Proportion-of-Failures Test

Training window: Jan 2019 – Dec 2023 (1,256 days). Test window: Jan–Dec 2024 (252 days). VaR was re-estimated using only training data; breaches were counted on the held-out test year.

Confidence	Method	Expected breaches	Actual breaches	Kupiec p-value	Verdict
95%	Historical	12.6	6	0.034	Rejected (over-conservative)
95%	Parametric	12.6	5	0.013	Rejected (over-conservative)
99%	Historical	2.5	0	0.024	Rejected (over-conservative)
99%	Parametric	2.5	2	**0.733**	**Not rejected**

Interpretation. The parametric 99% VaR was statistically well-calibrated on the 2024 hold-out — the model expected ~2.5 breaches, observed 2, and the Kupiec test cannot reject the null that the model is correctly calibrated. All three rejections are in the over-conservative direction — fewer breaches than expected, meaning the model over-estimated risk in 2024 rather than under-estimating it. This is a meaningfully different failure mode from an under-estimating model: an over-conservative VaR ties up excess risk capital but does not expose the firm to unexpected losses.

The most likely cause of the over-conservatism is that the training window (2019–2023) includes the high-volatility COVID and 2022 rate-shock periods, whereas 2024 was a comparatively calm year. A risk model trained on volatile history will over-state risk in calm regimes and vice versa. A production system would address this with a rolling-window or volatility-scaled re-estimation; this is outside the scope of this report but is called out as the natural next step.



5. Efficient Frontier & Optimization

Two approaches were used. A visualization was generated from 5,000 random Dirichlet-distributed weight combinations, plotting annualized return against annualized volatility coloured by Sharpe ratio. For the numerical optima reported below and used to construct Section 6's candidates, SLSQP constrained optimization was applied directly (constraints: weights ∈ [0, 1], sum to 1). The optimizer's values are cleaner and reproducible; the random-sampling visualization shows the shape of the feasible set.

Portfolio	Return	Volatility	Sharpe
Equal-weight (current)	16.71%	16.83%	0.725
Minimum variance	7.25%	10.14%	0.271
Maximum Sharpe	21.18%	17.19%	0.970

Candidate portfolio weights:

Ticker	Current	Min Variance	Max Sharpe
AAPL	12.5%	0.0%	43.5%

MSFT	12.5%	0.0%	3.9%
GOOGL	12.5%	0.0%	0.0%
JPM	12.5%	2.9%	6.0%
BRK-B	12.5%	20.3%	0.0%
GLD	12.5%	32.4%	46.6%
TLT	12.5%	35.7%	0.0%
SPY	12.5%	8.7%	0.0%

6. Portfolio Recommendation

Three candidate reallocations were evaluated against the equal-weight baseline, each targeting a different objective: maximizing risk-adjusted return, minimizing absolute loss exposure, and a balanced tilt that captures part of the optimization gain while preserving diversification. All three were derived from SLSQP-constrained mean-variance optimization over the 2019–2024 period. Together they span the relevant decision space along the efficient frontier. Each is presented below with its full risk-return trade-off; the recommendation follows in Section 6.4.

6.1 Candidate A — Maximum Sharpe (unconstrained mean-variance optimum)

Concentrated in AAPL (43.5%) and GLD (46.6%), with minority allocations to JPM (6.0%) and MSFT (3.9%) and zero weight in GOOGL, BRK-B, SPY, and TLT.

Metric	Baseline	Max Sharpe	Change
Expected return	16.71%	21.18%	+447 bps
Annualized volatility	16.83%	17.19%	+36 bps
Sharpe ratio	0.725	**0.970**	+0.245
99% Historical VaR	$29,146	$29,872	+2.5%
99% CVaR	$43,829	$40,739	–7.0%

This allocation is mathematically optimal in single-period mean-variance terms and establishes the theoretical upper bound on Sharpe improvement available from the 8-asset universe. However, the 90% combined weight in AAPL and GLD represents material single-name and single-theme concentration risk. A production risk committee would typically apply position caps (10–25% per name is a common range) that would rule this allocation out as implemented. It is presented here as the analytically valid upper bound, not as an operationally implementable recommendation.

6.2 Candidate B — Moderate tilt (blended allocation)

A 70%-equal-weight plus 30%-max-Sharpe blend. This produces small, interpretable shifts from the baseline: AAPL and GLD rise from 12.5% to approximately 22%, the remaining six assets hold weights between 8.8% and 10.5%, and no position drops below 8.8% or exceeds 23%.

Metric	Baseline	Moderate Tilt	Change
Expected return	16.71%	**18.05%**	+134 bps
Annualized volatility	16.83%	**16.45%**	–38 bps
Sharpe ratio	0.725	**0.823**	+0.098
99% Historical VaR	$29,146	**$29,032**	–**0.4%**
99% CVaR	$43,829	$42,318	–3.4%

The moderate tilt **strictly dominates the baseline on all four measured dimensions**: higher expected return, lower volatility, higher Sharpe, lower VaR. This is the allocation recommended as the principal action, for the reasons developed in Section 6.4.

6.3 Candidate C — Minimum variance (defensive mandate)

Heavily weighted toward fixed income and non-equity assets: TLT (35.7%), GLD (32.4%), BRK-B (20.3%), with a minority position in SPY (8.7%) and a token JPM allocation (2.9%). Zero weight in the three named tech positions and GOOGL.

Metric	Baseline	Min Variance	Change
Expected return	16.71%	7.25%	–946 bps
Annualized volatility	16.83%	**10.14%**	–669 bps
Sharpe ratio	0.725	0.271	–0.454
99% Historical VaR	$29,146	**$15,507**	**–46.8%**
99% CVaR	$43,829	**$22,839**	**–47.9%**

This portfolio achieves the maximum tail-risk reduction available in the feasible set — 99% VaR nearly halved, 99% CVaR nearly halved. However, the Sharpe ratio falls materially because the reduction in expected return outpaces the reduction in volatility: the portfolio is less efficient per unit of risk, even while being less risky in absolute terms. This allocation is appropriate under two distinct mandates: (a) a capital-preservation objective — near-retirement portfolios, endowment drawdown phases — where absolute loss matters more than risk-adjusted return, and (b) a regulatory VaR-minimization objective where an institution is required to hold the minimum feasible VaR under capital rules.

6.4 Principal Recommendation

Implement Candidate B (moderate tilt) as the primary reallocation. Three considerations support this decision:

1. **It strictly dominates the baseline on every measured metric.** Return, volatility, Sharpe, and VaR all improve simultaneously. This is unusual in mean-variance space and is not true of either alternative — Candidate A raises VaR while improving Sharpe; Candidate C lowers VaR while collapsing Sharpe. Candidate B is the only allocation that improves both absolute risk and risk-adjusted return relative to baseline.
2. **It is implementable.** No single name exceeds 23%, every asset retains meaningful exposure, and the weight changes are small — at most a ~10-percentage-point shift. This level of turnover is achievable in a single rebalancing with modest transaction costs, unlike Candidate A's 30-point concentration shift.
3. **It is consistent with the original portfolio thesis.** The equal-weight construction was motivated by diversification across three asset classes (equities, bonds, commodities). Candidate A abandons that thesis by eliminating exposure to four of the eight assets. Candidate C abandons equity exposure entirely. Candidate B is the only option that preserves the diversification narrative while still capturing meaningful optimization gain.

Candidate B captures approximately 40% of the Sharpe improvement available from the unconstrained Candidate A (0.098 of the 0.245 point gap) while maintaining the structural diversification that makes the portfolio defensible to a risk committee.

6.5 When the alternatives become correct

Candidates A and C are not inferior analyses — they are correct answers to different questions. This section documents when each would become the recommended choice, so that this report's framework is reusable under different mandates.

Candidate A (Max Sharpe) is the correct recommendation when:

- Single-name concentration caps are relaxed above 50%
- The mandate explicitly prioritizes return-maximization over diversification
- The investor can tolerate material company-specific risk (e.g., a concentrated family office or an active single-manager fund)

Candidate C (Min Variance) is the correct recommendation when:

- The primary objective is capital preservation rather than risk-adjusted return (e.g., near-retirement drawdown, insurance reserve portfolio)
- Regulatory VaR minimization is the binding constraint (e.g., capital-constrained bank prop desk)
- The investor's time horizon is short enough that Sharpe-maximizing compounding is secondary to avoiding a peak-to-trough loss

6.6 Comparison table — all four portfolios

Metric	Baseline	Max Sharpe		Min Variance
Expected return	16.71%	21.18%	**18.05%**	7.25%
Annualized volatility	16.83%	17.19%	**16.45%**	10.14%
Sharpe ratio	0.725	0.970	**0.823**	0.271
99% Historical VaR	$29,146	$29,872	**$29,032**	$15,507
99% CVaR	$43,829	$40,739	**$42,318**	$22,839
Largest single-name weight	12.5%	46.6%	**22.7%**	35.7%
Non-zero positions	8	4	**8**	5
Dominates baseline?	—	No (VaR ↑)	**Yes (all 4)**	No (Sharpe ↓)

7. Limitations

1. **Historical window dependence.** All three VaR methods are calibrated on 2019–2024. If the future regime differs materially from this window, estimates will be off. The backtest directly shows this: a model trained on volatile history over-estimated risk in the calmer 2024 test year.
2. **Normal-distribution assumption in parametric and Monte Carlo.** Both methods assume returns are multivariate normal. The historical-vs-parametric gap of $5,147 at 99% is direct evidence this assumption understates tail risk. A production model would use a Student-t distribution or a historical-simulation bootstrap for Monte Carlo to better capture fat tails.
3. **Static covariance.** The rolling-correlation analysis in Section 2 shows correlations are not stable. Parametric and Monte Carlo VaR both use a single full-period covariance matrix, which misses regime-dependent correlation breakdowns — exactly the environments where diversification matters most.
4. **1-day horizon only.** Regulatory VaR is often computed at 10-day or longer horizons. Scaling 1-day VaR by $\sqrt{10}$ assumes IID returns, which is approximately — but not exactly — true in practice.
5. **No liquidity, transaction-cost, or tax modelling.** The Section 6 reallocation recommendations implicitly assume frictionless rebalancing. A production implementation would add turnover constraints, transaction-cost modelling, and (for taxable accounts) tax-lot-aware rebalancing.
6. **Single-period optimization.** Section 6's candidates are derived from a single-period mean-variance optimization. Multi-period or dynamic-programming approaches (Merton-style) would handle rebalancing frequency and horizon-dependent risk preferences more rigorously.